MIL-R-36563A
3 October 1969
SUPERSEDING
MIL-R-36563
8 July 1968

MILITARY SPECIFICATION

RESUSCITATOR, FIELD, 110 Volt, 60 Cycle, AC, or 24 Volt, DC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of mechanical, intermittent, positive, automatic resuscitator, suitable for ventilation of chemical warfare casualties as well as for general medical resuscitation, and for field use in either a contaminated or uncontaminated atmosphere.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

Military

MIL-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic and associated Electrical Equipment

FSC 6515

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MIL-R-3065	Rubber and Synthetic Rubber Compounds, General Purpose
MIL-A-5092	Adhesive, Rubber Base, General Purpose
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-H-7138	Hose Assembly, Oxygen-Breathing Connector to Regulator
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-D-10131	Discs, Inlet Valve
MIL-M-10156	Mounting Canister, Gas Mask Plates
MIL-P-15024	Plates, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment
MIL-M-38784	Manual, Technical, General Requirements for Preparation of
MIL-P-19834	Plates, Identification, Metal Foil, Adhesive Backed

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MS 16686	Medical Service Symbol
MS 22058	Connector, Oxygen Hose to Regulator

DRAWING

Defense Personnel Support Center (DPSC)

25014	Resuscitator, Field, 110 Volt, 60 Cycle, AC, or 24 Volt, DC (16 sheets) and Bill of Material

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other Publication. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Aerospace Material Specifications

AMS 2673	Aluminum Molten Flux (Dip) Brazing

(Application for copies should be addressed to Society of Automotive Engineers, Inc., 485 Lexington Ave., N. Y. C., N. Y. 10017.)

UNITED STATES OF AMERICA STANDARDS INSTITUTE (USASI)

USA B46.1	Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to United States of America Standards Institute, 10 East 40th Street, New York, N. Y. 10016.)

3. REQUIREMENTS

3.1 Material. Except as otherwise authorized by the Contracting Officer, material shall be as specified in the Bill of Materials accompanying the drawing and as specified herein. Material shall be free of all defects and imperfections that may affect the serviceability of the finished product.

3.2 Proprietary Components. When trade named items or other proprietary components are specified, products of other manufacturers may be substituted provided that they are equal to, and interchangeable with, the item specified, and written approval of the Contracting Officer has been obtained.

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3.3 Construction. The resuscitator shall be constructed in strict accordance with Defense Personnel Support Center (DPSC) Drawing No. 25014, revised, 3 October 1969, and the requirements described herein. Assembly of the components shall be accomplished in an area free of excessive dust and metal chips to assure that foreign matter does not enter the system during such assembly. Assembly in a shop area where metal fabrication occurs shall not be permitted. The resuscitator shall consist of the following components: (1) a casualty circuit, (2) a cycling unit, (3) an air compressor, and (4) designated accessories and repair parts, all housed within a plastic carrying case.

3.3.1 Casualty circuit. The casualty circuit shall consist of the component parts assembled as shown on sheet 5 of the drawing. Orientation of component parts shall be such that there is no twisting of hoses when the casualty circuit is in the operating position. Such orientation occurs when the long hose is led over the shoulder of a casualty placed on his back, and the inlet valve-canister mount assembly is clipped to the casualty's shirt or jacket one handbreadth below the chin, and the mask outlet of the non-rebreathing valve faces the casualty's mouth.

3.3.1.1 Hoses. Hoses shall be of the sizes specified on sheet 15 of the drawing. Hoses shall be fabricated from an ozone resistant rubber compound, wire reinforced, nylon stockinette covered, permanently marked in accordance with MIL-STD-130, and meeting the applicable portions of MIL-H-7138, for Type I Hose.

3.3.1.2 Inlet valve - Canister mount assembly. The long and short hoses shall be joined by means of the inlet valve-canister mount assembly. The ports of the neoprene canister adapter, part 80, shall be stiffened by means of pieces of black anodized aluminum alloy tubing, part 152, thereby presenting a rigid support for the hose clamps. A spring clip, part 158, shall be bolted to the base of the canister adapter, which has been reinforced by means of black anodized aluminum alloy stiffeners. The spring clip shall be in accordance with MS 22058, except that the yoke shall be omitted. The open end of the canister adapter shall be fitted with a gas mask canister mounting fabricated in accordance with MIL-M-10156, except that sheets 5 and 15 of DPSC Drawing No. 25014, revised 3 October 1969, shall be substituted for the Chemical Corps. drawings referenced in the specification. The rubber gasket, part 82, shall be formed from natural or synthetic rubber conforming to type RN509ABF or RS509ABF of MIL-R-3065. The

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adhesive used to cement the gasket in place shall conform to MIL-A-5092. The inlet valve disc, part 81, shall conform with MIL-D-10131, except that sheet 15 of DPSC Drawing 25014, revised 3 October 1969, shall be substituted for the Chemical Corps drawing referenced in the specification. The snap fasteners used to secure the inlet valve disc in place shall be as specified in the Bill of Materials accompanying DPSC Drawing 25014, revised 3 October 1969, and shall be made of brass, with a dull black oxidized finish. The gas mask canister mounting shall provide a leakproof connection for both a standard M-11 gas mask canister (see 3.4) and the black anodized aluminum alloy anesthesia hose adapter, part 89, fabricated in accordance with the drawing. The anesthesia hose adapter shall be fitted with a monel mesh metal screen filter as shown on the drawing.

3.3.1.3 Non-rebreathing valve. The non-rebreathing valve shall be clamped to the end of the short hose installed so as to maintain the alignment specified in 3.3.1, and to provide sufficient clearance for removal or assembly of the neoprene grommet on the valve shells. The valve components shall be fabricated as shown on sheet 14 of DPSC Drawing No. 25014, revised 3 October 1969. Assembly shall be in accordance with sheet 5 of the drawing. The assembled non-rebreathing valve shall pass the performance tests specified in paragraphs 3.9.2 and 3.9.5 (f) and (g).

3.3.1.3.1 Non-rebreathing valve, alternate construction. As an alternate, the valve body and the upper and lower valve shells may be one-piece aluminum alloy die castings produced by the cold chamber process. Slight variations in design will be permitted to optimize construction, provided that the critical dimensions are maintained and the valve meets the performance tests. The alloy used as well as design shall be approved by the Contracting Officer prior to fabrication of parts.

3.3.1.4 Hose adapter. The hose adapter, part 86, shall be clamped to the end of the long hose. The black anodized adapter shall be fabricated as shown on sheet 15 of the drawing. An "O" ring, size 11, shall be fitted into the adapter groove thereby permitting a leakproof connection of the casualty circuit to both the mouthpiece (3.3.5.3) and the cycling unit outlet port (3.3.2), as applicable.

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3.3.1.5 Marking. The lower shell of the non-rebreathing valve and the anesthesia hose adapter shall be marked to identify ports by functions as shown on the drawing. Marking shall be by means of metal foil adhesive backed identification plates (see 3.3.6) or by engraving directly on the aluminum. If engraved, the characters shall be filled with a white lacquer after anodizing or hard coating. In addition, the upper part of the rubber non-breathing valve upper seat and the inner surface of the upper valve shell shall be color-coded (as shown on sheet 5 of the drawing) by means of a dot of red paint, as an aid to valve assembly. The paint shall have such adhesion to the neoprene so as to avoid chipping off as a result of normal valve flexure and handling.

3.3.2 Cycling Unit. The cycling unit shall be fabricated of detail parts shown on sheets 9, 10 and 11 and assembed as shown on sheet 3 of the DPSC Drawing No. 25014, revised 3 October 1969. In addition, the pressure gauge adapter shall be fitted with a wire snubber, 177, as shown on sheet 1 of the drawing.

3.3.2.1 Pneumatic circuit. The pneumatic circuit of the cycling unit shall be as shown on sheet 3 of the drawing. Pneumatic components shall be as specified in the Bill of Material accompanying the drawing and as described herein. The pneumatic circuit shall drive the bellows unit in a predetermined fashion in accordance with the parameters specified in paragraph 3.3.2.1.5. The pneumatic circuit shall consist of a filter unit, providing automatic water drainage, a pressure regulator, 104, (for adjustment see 3.3.2.1.5.2), a three-way valve, 107, three (3) bleeder valves, 113, (actuated as specified in 3.3.2.1.4), a flow control valve, 109, an actuating cylinder, 25, (3.3.2.1.1), a quick discharge valve, 22-24, (3.3.2.1.3), a needle valve, 111, and a muffler, 112. Physical arrangement of the components and mounting arrangements shall be as shown on sheet 3 of the drawing. Components shall be interconnected by means of the fittings and 1/8 inch diameter nylon tubing specified in the drawing. Teflon tape, 1/4 inch wide, shall be used to seal all pipe connections. Extreme care shall be taken on making joints to prevent shreds of teflon tape from entering into the system. The three (3) bleeder valves and elbow fittings shall be pretested for leakage (see 3.9.3) prior to installation in the circuit. Bleeder valves having leaks may be resealed and retested, and, if acceptable, may be installed.

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3.3.2.1.1 Actuating cylinder. Cylinder shall be Bimba Manufacturing Co., model D-1520-A or equal having the following salient characteristics: The single acting, spring return actuating cylinder shall be concentrically mounted within the bellows unit (3.3.2.1.2). The physical configuration of the cylinder shall be as shown on sheet 9 of the drawing. The cylinder shall be a 1-1/2 inch diameter bore by a 2-5/8 inch stroke. The body shall be fabricated from AISI type 304 stainless steel polished tubing which is crimped to anodized aluminum alloy ends. The piston shall be fitted with a Buna N, "U" cup seal. The piston rod shall be polished stainless steel rod which in the retracted position is flush to +1/64 inch above the cylinder end. The enclosed return springs shall exert a force of 6 to 8 pounds retracted, 12 to 14 pounds extended. The cylinder shall be packed with Dow Corning No. 55 pneumatic silicone grease or equal. The cylinder shall be rated for 250 psi air service.

* 3.3.2.1.2 Bellows unit. The bellows unit shall be assembed as shown on sheet 10 of the drawing. Particular care shall be taken to orient the upper and lower bellows plates so that the pressure tap (center two holes in series of four holes) on the upper plate are in proper angular relation to the upper bleeder valve spring loaded striker tab securing angle bracket (secured at the two tapped holes) on the lower plate. A suggested means of keying the two plates (semi-circular cut-outs) is shown on the detailed drawings. Alternate keying methods are permitted provided that excessive material is not removed which would provide inadequate gasket sealing surface when the bellows unit is connected into the cycling unit sub-assembly. A sector of the lower bellows plate and sealing ring shall be painted with a fluorescent paint, in the location shown on the drawing, using the paint specified in the Bill of Material.

* 3.3.2.1.3 Quick discharge valve. The quick discharge valve shall serve to by-pass a quantity of actuating cylinder driving gas around the needle valve, part 111, which controls expiration time thereby permitting a rapid acceleration of the bellows at the start of the expiration phase of the bellows travel. The negative pulse thereby generated in the bellows shall serve to actuate the non-breathing valve permitting unimpeded exhalation by the casualty to the atmosphere. The quick discharge valve shall be fabricated of detail parts shown on sheet 9 of the drawing, assembled as shown on sheet 3 of the drawing, coordinate B-5. The valve shuttle, part 24, shall be fitted with a No. 5 Quad seal, part 120, and the shuttle shall be lubricated with Dow Corning No. 44 light consistency silicone grease, or

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equal prior to insertion into the valve housing, part 23. In addition, the interior of the valve housing shall be coated with the silicone grease prior to insertion of the shuttle. Finally, the valve fitting, part 22, with No. 5 "O" ring in seal in place shall be inserted into the valve housing and secured by means of three screws. The physical location and inter connection of the quick discharge valve with other cycling unit components shall be as shown on sheet 3 of the drawing. For leakage characteristics of the quick discharge valve see 3.9.4. The valve capacity shall be such as to obtain the desired minimum obstruction frequency (3.3.2.1.5.7) to permit rapid exhalation via the non-rebreathing valve when the cycling unit volumn cycles, see 3.9.5(g).

3.3.2.1.4 Bleeder valve actuation. Two (2) of the bleeder valves shall be actuated by the moving bellows at stroke extremes (volume control), the third bleeder valve shall be actuated by a biased diaphragm sensitive to the pressure within the bellows (pressure control). Normally the bellows shall be volume cycled between the two volume controlling bleeder valves. However, once the pressure in the bellows reaches a predetermined value (see 3.3.2.1.5.7), the bellows shall be cycled at a reduced stroke by an increase frequency (pressure control) so that a fairly constant amount of ventilating gas is furnished to a casualty in a given time interval regardless of casualty impedance. The type of control, volume or pressure, shall be automatically determined by the impedance to ventilation offered by the casualty at that particular time.

3.3.2.1.4.1 Volume control. The two bleeder valves shall be actuated by the bellows unit at the extremes of its travel. The lower bleeder valve shall be actuated directly by the lower bellows plate. The upper bleeder valve shall be actuated by a spring loaded striker tab. The physical locations of the bleeder valves shall be set as specified in 3.3.2.1.5.5. The spring loaded striker shall serve to trip the upper bleeder valve for a longer period time (i.e. shall act as a time delay by having the spring loaded tab maintain contact with the bleeder valve actuating button despite start of downward movement of the bellows) thereby permitting a greater discharge of driving gas from the actuating cylinder via the quick discharge valve on start of expiration when the unit is volume cycled than when it is pressure cycled. The greater discharge shall cause a snap action of the non-rebreathing valve rubber valve elements to the open position thereby permitting unimpeded exhalation by the casualty to the atmosphere.

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3.3.2.1.4.2 Pressure control. A third bleeder valve shall be actuated by means of a spring biased diaphragm arrangement which is sensitive to the pressure of the ventilating gas generated in the bellows. Once the pressure in the bellows exceeds a predetermined valve, the third bleeder valve shall be actuated in lieu of the volume control upper bleeder valve so that bellows cycling occurs between the third bleeder valve and the lower bleeder valve of the volume control set. The pressure control sub-assembly shall be as shown on sheet 3 of the drawing, coordinate F-3. When installing the component parts, detailed on sheets 10 and 11 of the drawing, the parts shall be properly aligned to provide a clear pressure tap from the bellows, via the two holes, through the lower and upper diaphragm rings, to the top surface of the diaphragm. It is essential that the two rings shall not be interchanged in physical position. When installing the diaphragm subassembly consisting of the diaphragm, part 30, diaphragm stiffeners, parts 40 and 41, and striker rod, part 46, between the two rings sufficient slack shall be permitted when the securing screws are tightened to permit free vertical movement of the diaphragm striker rod in response to pressure changes on the upper surface of the diaphragm. The physical location of the bleeder valve with respect to the diaphragm striker rod shall be set to obtain the gap shown on the drawing. The gap shall be set with no gas pressure acting on top of the diaphragm,, and with the pressure adjusting screw set in approximate mid-position. The setting of the pressure control shall be as specified in 3.3.2.1.5.7.

3.3.2.1.5 Operating parameters. All controls of the cycling unit shall be pre-adjusted and locked in place to provide the specified mechanical operating parameters. The cycling unit shall be tested for compliance with specified settings in performance tests specified in paragraph 3.9.5.

3.3.2.1.5.1 Power Source. When setting the controls, the basic source of power shall be a clean, dry compressed air source having an adequate air capacity to maintain a minimum primary pressure of 30 pounds per square inch gauge (psig) at the cycling unit pressure regulator input port.

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3.3.2.1.5.2 Pressure regulator adjustment. The cycling unit pressure regulator shall be set to obtain a secondary pressure of 28.0(+2.0,-1.0) psig. The regulator shall be set under no flow conditions on a bench set-up. Alternately, the regulator may be set while in place on the cycling unit by attaching an accurate pressure gauge to the regulator gauge port, or on the secondary outlet tube, see 4.3.5(a), and setting regulator on the down stroke of the bellows or with secondary port blocked. The adjusting screw of the regulator shall be backed off before applying primary air pressure, and then the screw adjusted for the desired setting. Upon completion of adjustment, the pressure gauge shall be removed and, if required, the pressure gauge port shall be securely plugged.

3.3.2.1.5.3 Casualty load. When setting the cycling unit controls, the cycling unit shall have connected a properly functioning casualty circuit (paragraph 3.3.1) to which is connected a resistive load, or a resistive-capacitive load, as applicable, and as specified below.

3.3.2.1.5.3.1 Resistive load. When setting the inspiration and expiration times, the load shall consist of a resistor (an orifice) at the non-rebreathing valve mask port equivalent to a maximum of 20-25 centimeters (cm) H_2O pressure drop on inspiration, as recorded by the cycling unit pressure gauge (properly set at zero), when the cycling unit is operating in accordance with the parameters specified below. The Contractor shall make available an aluminum alloy, or other metal, test resistive load, fitted at one end with a male tapered joint to match the inside taper of the non-rebreathing valve mask port. The other end of the load shall conveniently hook to a capacitive load, see 3.3.2.1.5.3.2, when required, by means of rubber tubing, or other leakproof connection. A load which measures 2-1/4 inches long, and has a through 0.173 inch diameter (No. 17 drill) longitudinal hole will ordinarily provide a pressure drop within the specified range. Slight variation in length and orifice diameter is authorized in order to obtain a resistive load having the desired pressure drop.

3.3.2.1.5.3.2 Capacitive load. When setting the pressure control, in addition to the resistive load specified above, a capacitive load in series with this load shall be employed. The capacitive load shall consist of a rigid container, of approximately one gallon capacity simulating a low

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compliance casualty. A larger container may be employed provided it is filled with water, or other inert material, to provide a capacity, which in series with the resistor specified, causes the cycling unit to assume pressure cycling, when the cycling unit is operating in accordance with the parameters specified below. Aside from causing pressure cycling, the absolute value of the capacitance (or compliance) shall not be significant.

3.3.2.1.5.4 Ambient temperature. When setting the controls of the cycling unit, the room ambient temperature shall be maintained at 65° to 75° Farenheit (F).

3.3.2.1.5.5 Bellows volume output. The bellow volume output shall be set by the physical spacing between the upper and lower bleeder valves to a value of 900 (-100, +0) cubic centimeters (cc). The output shall be measured by means of an anesthesia ventilation meter, such as a Wright Respirometer, A Monoghan Bennett Ventilation Meter, or a Drager IM-2470 meter. The meter shall be connected to the mask output port of the non-rebreathing valve in accordance with meter manufacturer's instructions. In setting the bleeder valves, the lower bleeder valve shall be set and locked first by pushing it up against the lower bellows plate (with the actuating cylinder rod fully retracted) until the bleeder valve button is depressed. The upper bleeder valve shall then be set and locked in place to obtain the specified bellows volume output, while ensuring positive action of the bleeder valve actuating button within the stroke range of the actuating cylinder and prior to reaching the stacked height of the rubber bellows. The setting shall be in room ambient temperature specified in 3.3.2.1.5.4.

3.3.2.1.5.6 Inspiration-Expiration Times. The inspiration flow control valve and the expiration needle valve shall be set and locked to maintain the following times:

Inspiratory period	$1.25 \begin{smallmatrix} +0.10 \\ -0.0 \end{smallmatrix}$ sec.
Expiratory period	$2.50 \begin{smallmatrix} +.00 \\ -.10 \end{smallmatrix}$ sec.
Frequency (reference only)	16 cycles/minute

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During setting the resuscitator the ambient temperature shall be as specified in 3.3.2.1.5.4. The timer shall be of an electric stopwatch type, capable of timing in .01 of a second. Time shall be recorded between the actuation of the bleeder valve buttons. An average of a minimum of three (3) readings shall be used as an indication of the time setting. Alternately, an electronic recorder capable of recording instantaneous pressure or flow conditions against time (by means of suitable sensors such as strain gages) may be used to set the controls. The time scale shall be capable of expansion so that intervals .01 of a second can be read.

3.3.2.1.5.7 Pressure Setting. The pressure setting shall be accomplished with a resistive-capacitance load specified in paragraph 3.3.2.1.5.3.2, and at a room ambient temperature as specified in 3.3.2.1.5 4. The pressure adjusting screw (part 45) shall set the spring bias on the diaphragm so that volume cycling occurs prior to complete excursion of the bellows, when the pressure recorded by the cycling unit pressure gauge reads 60±2 cm. H_2O pressure. Before making this setting, the bleeder valve must be properly positioned and locked in place with respect to the diaphragm striker rod (see 3.3.2.1.4.2). The bias on the diaphragm shall be set by means of the pressure adjusting screw so that the specified bellows pressure actuates the bleeder valve. It is essential that a slight gap shall exist between the diaphragm striker rod and the bleeder valve acuating button when no pressure exists in the bellows. In addition, the gap shall not be so large as to require excessive travel of the diaphragm striker rod to actuate the bleeder valve. If necessary, the gap may be reset, by trial and error, to obtain sensitive actuation. Once the adjusting screw is properly set it shall be locked in place by means of the jam nut. On pressure cycling, the pressure within the bellows shall fall rapidly from its preset valve (60cm H_2O) to a maximum of 30 cm H_2O (due to the action of the quick discharge valve) before dropping at a slower rate to zero. Due to the incomplete instantaneous pressure drop, there shall be a resonating of rubber valve elements in the nonrebreathing valve, the resonating noise serving as an audible indication that the casualty is highly impeded and requires additional medical attention. The pressure setting shall be checked by removing the resistive-capacitive load and completely obstructing the non-rebreathing valve mask port to obtain the obstruction signal. The signal shall consist of a rapid cycling of the bellows, at a minimum frequency of 70 cycles per minute and at approximately the same maximum pressure as previously specified.

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3.3.3 Air Compressor Sub-Assembly. The air compressor sub-assembly shall consist of an air-compressor, an electric motor, a transformer-rectifier power source, electric cords, pulley and belt drive and necessary mounting sheet metal components. Detail parts shall be fabricated as shown on sheets 12 and 13 of the drawing. Assembly shall be as shown on sheet 4 of the drawing. Components shall be mounted within the tolerances specified in the drawing to assure proper fit of the unit within the plastic carrying case. For performance of air compressor see 3.9.7.

3.3.3.1 Compressor. Unit shall be K. J. Miller Corp., Cat. No. 2023, air compressor, or equal with the following salient characteristics: The compressor shall be of the style and design shown on the drawing. It shall consist of a cast iron, normalized and precision machined and roller burnished cyclinder provided with a 1 inch bore, so machined to meet the diametrical clearance with the piston specified below. Cylinder shall be finned to promote heat dissipation. The piston shall be an aluminum die casting precision fitted within the cylinder to maintain a maximum diametrical clearance of 0.001 inch. Piston rings or seals shall not be required. Piston shall be wick lubricated (see 3.3.3.1.1), the piston making contact with a ring-shaped felt wick on the extreme downward stroke. The head of the piston shall be provided with the inlet valve consisting of a phenolic disk, working on a brass valve seat, held in place by means of a steel retainer spun in place. The piston connecting rod shall be a die-casting provided with lubricant impregnated sintered bearings at each end. The connecting rod shall be attached to the piston by means of a 1/4 inch diameter precision ground wrist pin. The crankcase shall be a zinc die casting. Inlet air shall enter a rectangular port in the crankcase which is fitted with a felt muffler cemented in place. The journal shall be secured to the crankcase by means of four screws and shall be fitted with two 1/2 inch sections of sintered lubricant impregnated bearings. An oil hole fitted with a felt plug shall provide access of lubricant to the bearing sections. An additional oil hole in the crankcase shall provide lubricant access to the piston ring-shaped felt wick. The crankcase shall be provided with four tapped (10-32 tap) holes in the base conforming to the mounting dimensions shown on sheet 12 of the drawing for part 56. A cork crankcase shall be provided. The compressor shaft shall be 3/8 inch diameter precision ground steel. A counterbalanced crank shall be provided. Shafts shall be secured to crank by means of a press fit and

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then staked in place under high pressure. The shaft shall be provided with a flat section to receive the set screw of the compressor pulley (Part 65). The cast iron cylinder shall be secured to the crankcase by means of four screws. The exhaust valve shall be a mylar sheet working on a stainless steel valve plate. No springs shall be required for proper functioning of the exhaust valve. The outlet air dome shall be a finned aluminum die casting and shall be provided with two 1/8 inch female pipe thread outlets, at 180 degrees to each other, and at right angles to the compressor shaft. The outlet air dome, outlet valve, valve plate and gaskets shall be secured to the cylinder by means of four screws Sealing shall be by means of graphite gaskets. The compressor shall have a maximum weight of 2-3/8 pounds. The capacity of the compressor shall be such that when it is operated at a speed of 2100 revolutions per minute (R. P. M.) into the load specified in 4. 4. 6 (see 3. 9. 6), the tank pressure gauge of the designated load shall read 15 psi minimum. All compressors shall be checked by the compressor manufacturer for performance prior to use in the resuscitator air compressor unit. Compressor shall be finished in a commercial paint finish.

3. 3. 3. 1. 1 Lubricant. The compressor shall be lubricated with Anderol L-465 synthetic impregnating oil, Lehigh Chemical Company, Chestertown, Maryland, or equal.

3. 3. 3. 1. 2 Compressor fitting. The compressor shall be fitted with relief valve (part 127) in one port, and a street elbow (part 110) and quick-connect coupling (part 101) at the other port as shown in the assembly on sheet 4 of the drawing. The relief valve shall be set to crack at 30 psi, within the commercial tolerances specified by the manufacturer for the designated relief valve.

3. 3. 3. 2 Electric Motor. A 24-29 volt direct current electric motor shall drive the compressor by means of a positive drive belt. The motor shall be shunt wound, continuous duty, rated at 0. 075 H. P., 4200 R. P. M. The motor shall comply with the dimensions for part 67, sheet 13 of the drawing. The motor shall be provided with radio interference reduction feed through capacitors (see 3. 3. 3. 5). The motor shall be equipped with a rear external fan and fan shroud so that when the motor is operated under load at a speed of 3800 R. P. M. the air velocity through the motor shall be approximately 200 feet per minute, and air velocity over the motor shall be approximately 2000 feet per minute. The shroud shall be provided with a

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plastic shield(part 62) to protect the feed through capacitors. The motor shall be designed to have a minimum life of 5000 hours. The designed brush life shall be a minimum of 1000 hours. The motor shall be provided with permanently lubricated anti-friction bearings. The commutator design shall be such as to minimize sparking. The motor shall be anti-fungus treated as specified in MIL-T-152, and finished to meet the requirements of MIL-E-5272. The weight of the motor (part 67) shall not exceed 4.1 pounds. The motor shaft shall be provided with a flat to receive the set screws of the modified pulley (part 66). Feed through capacitor and brush holder shall be provided with polarity markings as specified in the drawing.

3.3.3.3 Transformer-Rectifier Power Source. A transformer-rectifier power source shall be provided to permit operation of the direct-current (D.C.) motor from a 105-130 volt alternating current (A.C.), 60 cycle power source. The source shall consist of essentially of a step-down transformer, a silicon diode full wave bridge rectifier, rectifier heat sinks, a switch, fuses with holders and power cords with the specified plugs and receptacles. A diode shall be provided to prevent operation of the motor from D.C. power of incorrect polarity. The wiring shall be in accordance with the diagram shown on sheet 4 of the drawing. The components, the physical location and the mounting of components shall be as specified in the drawing. Silicon compound shall be used to insure good thermal contact between the diodes and the heat sinks.

3.3.3.3.1 Transformer. The transformer shall be the Hall Manufacturing Co. S-252-X transformer or equal with the following salient characteristics: The transformer shall be a compact lightweight toroidal wound step-down unit. When a primary voltage of 115 volts, 60 cycle A.C. is applied, the secondary open circuit (no load) voltage shall be 34.6 volts; the full load secondary voltage shall be approximately 33.0 volts. The transformer core shall be Magnetics, Inc. 50022-12K. BM-12KG material or equal. The primary shall consist of 1230 turns of No. 22 heavy Formvar wire, the secondary of 370 turns of No. 18 heavy Formar wire. Primary leads shall be color-coded black, secondary leads shall be color-coded green. No. 27 Minnesota Mining & Manufacturing Co. tape or equal shall be used between windings and as a final winding. The unit shall be vacuum impregnated with Minnesota Mining and Manufacturing Co.'s No.

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280 clear epoxy compound, or equal and then baked for 12 hours at 210°F to meet Class F. service. The maximum weight of the transformer shall be 2.7 pounds. Nominal dimensions shall be 1-3/4 inches high, 3-7/8 inches outside diameter, 7/8 inches inside diameter.

3.3.3.3.2 Power cords. Two electric power cords shall be provided. The A.C. cord shall comply with requirements for part 68 and the D. C. cord for part 69 of the drawing. When shipped the A.C. cord shall be installed and the D.C. cord stored in the storage space allotted (see 3.3.8).

3.3.3.4 Belt Drive. A positive drive belt, Goodyear Part No. 170XL025 or equal shall be used to connect the motor and compressor pulleys. Before installing the belt, the pulley should be aligned on the shafts. Shafts should be parallel and pulley alignment proved by means of a straight edge. Belt tension shall be adjusted by means of the two adjusting screws moving the motor mount hot section, guided on the two pushing guides part 143) in machined slots of the motor mount, toward the belt adjusting bracket (part 64). The belt should not be tightened exclusively, but should be installed with a snug fit. The belt tension should be checked with compressor under loaded and no-load conditions. A spare quick connect coupling insert (part 100) inserted into the quick connected coupling body (part 101) of the compressor shall be used to obtain no-load condition. Loaded condition shall be defined as that which results when the quick connect coupling insert is removed. Under these conditions, the belt tension shall be adjusted so that there is a compromise between vibration of the compressor unit and load (as measured by a wattmeter) drawn by the motor. After the belt has been adjusted and pulleys checked for alignment, the motor shall be tightened to the mount by means of the hex head bolts.

3.3.3.5 Radio Interference. The electrical system of the air compressor unit shall comply with the radiated and conducted electromagnetic emission limits prescribed in MIL-STD-461 for class IIB Equipment.

3.3.4 Carrying case. The plastic carrying case shall be formed or molded from material specified on the drawing as shown on sheets 2, 6, 7 and 8. The case body, cover, and base shall be fitted with mating aluminum

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alloy extruded valences, fitted with a tubular neophrene gasket cemented in place. The aluminum shall be clear anodized as specified in MIL-A-8625 for type II. Valences shall be sealed to the plastic by means of a clear silicone rubber sealer prior to being crimped into place. The case cover shall be hinged to the body by means of a continous hinge and shall be provided with two drawbolts for latching in the closed position. The case body shall be provided with a divider (providing a storage compartment), a clear plastic observation window, and a rear access door. The access door shall be hinged to the body by means of a continuous offset hinge and shall be latched closed by means of two snap fasteners mounted as shown on the drawing. A rubber gasket shall provide a rain-proof construction at the access door. The case body shall be provided with a carrying handle (positioned as shown) and four rubber bumpers. The body upper valence shall support the cyling unit and shall be provided with clearance cut-outs to facilitate installing and removal of this unit. The molded case base shall house the air compressor unit, the lower air compressor platform being secured to the case base at the four rubber bumpers, as shown on sheets 1 and 2 of the drawing. The base shall be joined to the body by means of four drawbolts. The inside of the cover of the case shall be provided with a spring clip (13) and plastic pad (170) for securing the aspirator, a plastic foam holder (15) for the mouthpiece, and a plastic pad (169) for protecting the pressure gauge during transit.

3.3.5 Accessories, Repair Parts. Each resuscitator unit shall be provided with the following accessories and repair parts. For storage parts, see 3.3.8.

3.3.5.1 Aspirator. The aspirator shall be constructed in strict accordance with sheet 16 of the drawing. The catheter used with the aspirator shall comply with the requirements specified in the current Military Medical Purchase Description for FSN 6515-317-4230 Catheter, Urethral; Rubber, Pointed Tip, Robinson, 14 Fr. (see 3.4). All joints shall be tight so that repeated squeezing of the bulb shall induce sufficient negative pressure in the jar to draw in fluid into which the end of the catheter is immersed.

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3.3.5.2 Airways. A large adult and small adult airway shall be furnished. The airways shall comply with the requirements specified in the current Military Medical Purchase Descriptions for FSN 6515-300-2900 Airways, Pharyngeal, Rubber, Large Adult and FSN 6515-300-2910 Airway, Pharyngeal, Rubber, Small Adult (see 3.4).

3.3.5.3 Mouthpiece. The mouthpiece shall be fabricated from aluminum alloy tubing as detailed (part 85) in the drawing. The outer surface of the mouthpiece shall be coated using an aerated fluidized bed, or other method, with a smooth layer, approximately 1/32 inch thick, of polyethylene, Union Carbide DKPB-0850 or DKPC-0850 natural color resin or equal. As an alternate, the exterior of the mouthpiece may be coated with an acrylic white, 2 coats, each having a minimum thickness of 1 mil. The coating shall be applied by Technicote Division, L.J. Kissling & Son, L.I.C., N.Y. and the acrylic shall be formula X-RT14. The interior of the mouthpiece shall be reamed to make a leakproof connection with the hose adapter, part 86.

3.3.5.4 Mask. The apparatus shall be equipped with an electrically conductive mask, medium adult, anotomical type, with inflatable rim. The mask shall be constructed of molded rubber and stiffened with metallic inserts to permit and retain minor reshaping when desired. The mask shall be equipped with a suitable hook to permit the attachment of a head strap (3.3.5.5.) to maintain the mask firmly upon the face of the patient. A molded conductive rubber backing shall be provided for attachment to the non-breathing valve. The mask shall comply with the current Military Medical Purchase Description for FSN 6515-347-2200 Mask, Oronasal, Resuscitator and Anesthesia, Medium Adult (see 3.4).

3.3.5.5 Head Strap (Retainer). The head strap, part 165, shall be fabricated from an electrically conductive rubber. Unit shall be the No. 4 Inhaler Retainer, Ohio Medical Products, Part No. 211-1676-100 or equal.

3.3.5.6 Repair Parts, Rubber Goods. The following rubber repair parts shall be packed in a plastic bag (part 122):

(a) Gasket, Valve Body (Part 74)
(b) Valve, Exhaust (Part 75)
(c) Seat, Non-Rebreathing Valve (Part 76)
(d) Disk, Non-Rebreathing Valve (Part 77) assembed on seat (c) above.
(e) Grommet, Valve Sheel (Part 78)

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In addition, 1 ounce of silicon grease, Dow Corning No. 44, or equal light consistency in a suitable collapsible tube or in a clear plastic vial (part 121) shall be packed within the spare parts bag.

3.3.5.7 Repair Parts, Electrical. The following electrical repair parts shall be packed in a plastic bag (part 122):

(a) Belt, Positive Drive (Part 128)
(b) Fuses, 1 6/10 Amp, Box of 5 (Part 133)
(c) Fuses, 6 1/4 Amp, Box of 5 (Part 134)
(d) One Set of Motor Brushes, consisting of a Positive and Negative Brush (Part 238). Brushes shall be packed in a small rigid container similar in size to a fuse box.

(e) One plug cap, Hubbell No. 7545 or equal. A tag shall be attached to plug stating that plug cap be substituted for the battery clips on the D.C. cord for use in a vehicle equipped with a mating electrical outlet.
(f) One connector (FSN 5935-991-0004). A similar tag as for (e) above shall be provided.

3.3.5.8 Lubricating Oil. A 1 ounce can, with spout, of lubricating oil, Anderol L-465, or equal, part 141, shall be packed within a plastic bag (Part 123).

3.3.5.9 Gas Mask Canister. The gas mask canister shall comply with current requirements for FSN 4240-112-9365 Canister, Field, Protective Mask, M-11. The canister shall be Government furnished material (see 3.4).

3.3.5.10 Repair fittings. The following repair fittings shall be provided:

(a) Bushing, 1/4 by 1/8 NPT (Part 173)
(b) Coupling, Quick Connect Body (Part 101)

3.3.6 Identification marking. The resuscitator unit shall be provided with identification plates, part numbers 9, 10, 11, 11A, 12, 52, 58, 59, 70, 90, 91, and 92 in positions indicated in the drawing. All plates except part 10, shall be adhesive backed metal foil and shall comply with MIL-P-19834, except that sizes shall be in accordance with the DPSC Drawing No.

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25014, revised 3 October 1969. Plate 10 shall comply with MIL-P-15024, for Type H, 0.032 inch thick aluminum, and shall be attached by adhesive as well as rivets. In addition, the carrying case shall be provided with a manufacturer's name plate (coordinates E-8 of sheet 1 of drawing) bearing the following information:

"Resuscitator, Field, 110 Volt, 60 Cycle, AC, or 24 volt, DC
FSN_____
Manufacturer's name or registered trademark
Model No._____
Unit Serial No._____
U.S. Patent No. 3,251,359
The medical services symbol, type III, Standard MS 16686"

Finally, the cycling unit shall be marked with the Unit Serial No. in the location designated on sheet 3 of the drawing (coordinate D-6). The manufacturer's nameplate shall comply with MIL-P-15024, for Type H, aluminum, and shall be applied by a suitable adhesive. The serial number may be applied to the cycling unit by means of an adhesive backed metal foil plate or by stamping or engraving directly onto the metal. In all instances, surfaces which will receive plates shall be properly prepared to insure adherance.

3.3.7 Finish.

3.3.7.1 Machine finish symbols. Machine finish symbols shall be as defined in USASI B46.1 - Surface Texture (Surface Roughness, Waviness and Lay).

3.3.7.2 Black anodizing. Aluminum alloy components which are specified to have a black anodized finish shall be finished in a heavy, uniform continuous black anodic film free of breaks, scratches or other damage. Surfaces which are visible shall be given a grained satin finish (approximately 280 grit) prior to anodizing. Anodic coating shall comply with MIL-A-8265, type II, black-dyed coating, class 2.

3.3.7.3 Hard coating. Aluminum alloy components which are specified to have a hard coating shall be finished to comply with MIL-A-8625, type III.

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3.3.7.4 Clear anodizing. Aluminum alloy components which are specified to have a clear anodized finish shall have a coating complying with MIL-A-8625, type II, class 1.

* 3.3.8 Assembly. The unit shall be assembled for shipment as shown on sheet 1 of the drawing. The compressor unit shall be mounted in the case base. The lubricating oil, within plastic bag, shall be shipped in the plastic foam oil can holder, part 61. The plastic bag containing the electrical repair parts (3.3.5.7) shall be placed in the case base, alongside the compressor so that it fits under the case storage compartment. The D.C. power cord shall be coiled and fitted on top of this plastic bag of electrical parts. The case base shall be secured to the body by means of the four drawbolts. The cycling unit shall be screwed into the upper case body valence by means of the six screws. The air line shall be coiled around the outlet manifold and fed through the grommetted hole in the cycling unit upper plate through the cut-out in the cycling unit lower plate and looped along the side of the case and snapped into the compressor quick connect fitting. The spare fittings (3.3.5.10) shall be screwed into the fitting mount (part 16) located on the inside of the case access door. The bag of rubber repair parts (3.3.5.6), the head strap (3.3.5.5), the gas mask canister (3.3.5.9), technical manuals (3.6), and the casualty circuit with mask attached shall be stowed in the case side storage compartment. The airways, mouthpiece and aspirator (with bulb and catheter stored in jar) shall be stowed in the placed provided in the case cover. The A.C. electric cord, attached to the compressor unit, shall be coiled and secured to prevent unwinding and stowed in the case body through the rear access door. The access door shall be closed and secured by the snap fasteners. The pressure gauge on the cycling unit shall be swiveled to a horizontal position and the case cover secured by the two drawbolts.

3.3.9 Weight. The finished weight of the resuscitator, within the carrying case, including accessories, repair parts, and manuals (3.6) shall not exceed 34 pounds.

3.4 Government Furnished Material. The Government shall furnish the contractor the following in sufficient quantity to provide one for each unit under contract:

FSN 4240-112-9365	Canister, Field, Protective Mask, M-11
FSN 5935-991-0004	Connector, Plug
FSN 6515-300-2900	Airway, Pharyngeal, Rubber, Large Adult

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FSN 6515-300-2910	Airway, Pharyngeal, Rubber, Small Adult
FSN 6515-317-4230	Catheter, Urethral, Rubber Tip, Robinson, 14 Fr.
FSN 6515-347-2200	Mask, Oronasal, Resuscitator and Anesthesia, Medium Adult.

No other Government furnished material will be provided.

3.4.1 Government loaned property. The following gages are available on a loan basis from the Defense Personnel Support Center:

Stock No. 5220-A50-6445 Go, Thread, Plug Gage
Stock No. 5220-A50-6459 Not Go, Flat Plug Gage.

These gages shall be used to check the Cannister Mounting Part No. 79. Upon completion of the contract, the gages are to be returned to the Defense Personnel Support Center, 2800 S. 20th Street, Philadelphia, Pa. 19101, Attn: Code ATTS-1.

3.5 Color. The color requirements shall be as specified in DPSC Drawing No. 25014, revised 3 October 1969.

* 3.6 Service data. The contractor shall enclose, with each unit, two copies each, of one or more service data booklets. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. The technical manuals shall be type I in accordance with MIL-M-38784, except approval shall be by the Defense Personnel Support Center. The service data shall contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number.

3.6.1 Approval. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order.

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3.6.2 Waiver of Review and Distribution Copies of Manual. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.7 Supplier responsibility for maintenance. The supplier shall, for a period of at least two years after the completion of contract, supply under contract or purchase order, such repair parts as are peculiar to his manufacture, or other commercial parts, modified by him to provide components for the resuscitator supplied by him.

* 3.8 Interchangeability. All removable component parts and assemblies shall be interchangeable with the respective parts and assemblies of all other resuscitators produced under this specification regardless of manufacturer, without adverse effect on the functioning of the end item.

3.9 Tests. The resuscitator shall perform in accordance with the following requirements:

3.9.1 Clinical tests. Unless otherwise specified, the Government will arrange for clinical testing of the resuscitator to insure that the unit functions properly to deliver safe and effective ventilation automatically to wide range of impeded casualties in accordance with the characteristics specified herein.

The contractor shall furnish instantaneous flow and pressure traces using simulated resistive and capacitive loads, to be approved by procuring activity. Traces shall be obtained as described in (4.4.1). Sample acceptable traces will be furnished to the contractor, upon request.

3.9.2 Leakage test, non-rebreathing valve. The non-rebreathing valve shall show no evidence of leakage when tested in accordance with procedure given in 4.4.2.

3.9.3 Leakage test, bleeder valves. Bleeder valves shall show no leakage of air at the bleed ports, or at the connection of the elbow fitting to the bleeder valve, when tested as described in 4.4.3.

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3.9.4 Leakage test, quick discharge valve. There shall be no leakage of air at the atmospheric port of the quick discharge valve during operation of the cycling unit, except when expiration is initiated by actuation of a bleeder valve, at which time there shall be a definite blast of air at this port, accompanied by an initial rapid acceleration of the bellows. Test shall be accomplished as described in 4.4.4.

3.9.5 Performance test, cycling unit and casualty circuit. When tested in accordance with 4.4.5, the cycling unit and casualty circuit shall be checked for:

(a) Pressure regulator adjustment (3.3.2.1.5.2)
(b) Bellows volume output (3.3.2.1.5.5)
(c) Inspiration-Expiration Times (3.3.2.1.5.6)
(d) Pressure setting (3.3.2.1.5.7)
(e) Obstruction frequency (3.3.2.1.5.7)
(f) Non-rebreathing valve resonation, pressure cycling (3.3.2.1.5.7)
(g) Non-rebreathing valve unimpeded exhalation, volume cycling (3.3.2.1.4.1)

3.9.6 Compressor capacity. The compressor capacity (3.3.3.1) shall be tested as described in 4.4.6.

3.9.7 Air compressor performance. When tested in accordance with procedure given in 4.4.7, the air compressor of the resuscitator shall operate a pre-tested cycling unit, over a range of A.C. voltages and on 24 volt D.C. power in such a manner so that the parameters previously measured (3.9.5(c), (d) and (e)) shall not vary by more than $\pm 10\%$.

3.9.8 Rain test, carrying case. There shall be a negligible leakage of water into the carrying case when the unit is subjected to the rain test specified in 4.4.9. After the test, the unit shall be in operating condition, meeting specified performance.

3.10 Workmanship. Workmanship shall be first class throughout. Resuscitators shall be free of defects which detract from their appearance or may impair their serviceability.

3.10.1 Welding aluminum. The welding shall be accomplished by inert gas shielded arc welding and shall be continuous, durable and sound in every detail, producing a leakproof connection of flow of air. Component members shall be carefully and properly aligned prior to welding into permanent position. When spotwelding is shown on drawing, such welds shall be free of surface burning, cracks, and porosity, and shall be tightly fused.

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3.10.2 Brazing aluminum. Brazing of aluminum shall be accomplished by immersion in a molten flux bath, i.e. by dip brazing. Brazing shall be accomplished in accordance with Aerospace Material Specification AMS 2673. Tack welding for fixturing will be permitted. Joints shall be continuous, durable and sound in every detail and shall offer a leakproof connection to flow of air.

3.10.3 Machine work. All component parts shall be free of burrs and other imperfections. All threads on threaded members shall be properly formed and of the fit specified in the drawing.

3.10.4 Assembly. All component parts of the cycling unit and particularly fittings, shall be aligned so that assembly and disassembly shall be easily accomplished without kinking or twisting of the nylon tubing (see sheet 3 of the drawing).

3.10.5 Solvent welding. Components on the plastic carrying case specified on the drawing as being joined by solvent welding shall be carefully aligned prior to joining. Joints shall be durable and sound in every detail. Solvent shall be carefully applied only to surfaces being joined. Gaskets, parts 225 and 226, shall be immersed in solvent, excess solvent squeezed out, and then carefully placed in position for final bonding to case.

3.10.6 Cleaning. After fabrication, and prior to finishing, metal parts shall be cleaned in accordance with good commercial practice. Cleaning processes shall have no deleterious effect on the equipment. After assembly, units shall be cleaned thoroughly to be free of any foreign material. Prior to packaging, the plastic carrying case shall be rubbed down with a wax-laden cloth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests other than those listed below shall be conducted in accordance with MIL-STD-105.

Sampling Criteria for Tests

Component	Characteristic	Rqmt.	Test procedure	Unit of product	Insp. Level	AQL
Non-rebreathing valve	Leakage	3.9.2	4.4.2	1-Casualty circuit	*	---
Bleeder valves	Leakage	3.9.3	4.4.3	1-Bleeder valve	*	---
Quick discharge valve	Leakage	3.9.4	4.4.4	1-Cycling unit	*	---
Cycling unit and casualty circuit	Performance	3.9.5	4.4.5	1-Resuscitator	*	---
Compressor	Capacity	3.9.6	4.4.6	1-Compressor	*	---
Compressor	Performance	3.9.7	4.4.7	1-Compressor	*	---
Resuscitator	Radio interference	3.3.3.5	4.4.8	1-Resuscitator	***	**
Resuscitator	Rain	3.9.8	4.4.9	1-Resuscitator	***	**

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* Each unit supplied shall be tested, and no defective units will be accepted.

** Acceptance number shall be zero for all sample sizes.

*** Radio interference and rain tests shall be performed on First Article samples only.

4.3 Examination. The resuscitator shall be examined to determine the materials, design, construction, physical dimensions, and workmanship are in accordance with applicable requirements. Visual and mechanical defects shall be classified during the course of the contract, by the procuring activity.

4.4 Test methods. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Instantaneous traces. Instantaneous flow and pressure traces shall be obtained by using simulated sets of impedances, each consisting of a resistive load and a capacitive load connected in series. Traces shall consist of the following pressures and flow vs. time (in seconds):

(a) Bellows pressure - cm H_2O
(b) "Lung" pressure - cm H_2O (measured in the capacitive load)
(c) "Mask" pressure - cm H_2O (measured at junction of non-rebreathing valve mask outlet and inlet to resistive load - see below)
(d) Ventilating gas flow - liters/sec. (measure at junction between resistive and capacitive loads)

To measure "mask" pressure it is recommended that a metal adapter having a center pressure tap be fabricated which has a taper joint at each one; one taper to match the outlet of non-rebreathing valve mask port, the other taper to match the resistive load. Pressures and flows shall be measured by means of suitable strain and flow gauges and recorded by means of a four channel electronic recorder, permitting simultaneous recordings of all traces. Two values of resistance shall be employed: (1) a low resistance of approximately 10 cm H_2O/liter per second, and (2) a high resistance of approximately 40 cm H_2O/liter per second, both resistance values being measured at 1.0 liter/second air flow. Two values of capacitance shall be employed: (1) a low compliance of approximately 0.015 liter/cm H_2O and (2) a high compliance of approximately 0.15 liter/cm H_2O. Four impedance sets, each consisting of a resistance and a capacitance shall be employed. In addition, a fifth set of traces, showing bellows pressure, mask pressure and ventilating gas flow vs. time shall be obtained for total obstruction.

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4.4.2 Leakage test, non-rebreathing valve. The casualty circuit shall be disconnected from the cycling unit and a mouthpiece, part 85, shall be installed. While blowing gently into the mouthpiece, the mask outlet port of the non-rebreathing valve shall be closed with a suitable stopper. An escape of air through the non-rebreathing valve exhaust port indicates leakage in the non-rebreathing valve which is not acceptable. If acceptable, shift the stopper to leave an opening of approximately 1/8 inch in diameter, or substitute a one-holed stopper in the mask outlet port, to permit passage of some air while blowing into mouthpiece. Air shall only escape through the small opening provided. There shall be no simultaneous escape of air through the exhaust port.

4.4.3 Leakage test, bleeder valves. The bleeder valve with associated male elbow, part 98, shall be connected by means of 1/8 inch nylon tubing to a compressed air source maintained at 20 psig. The bleeder, including fitting, shall be immersed in water. The presence of bubbles in the water indicating an air leak at the bleed ports, or at the fitting joint, indicates a non-acceptable sub-assembly. If acceptable, the water or unit shall be dried before installing into a cycling unit.

4.4.4 Leakage test, quick discharge valve. The test shall be accomplished on an assembled cycling unit which is being driven by a compressed gas source (3.3.2.1.5.1) and has been adjusted in accordance with specified operating parameters. The casualty circuit shall not be attached to the cycling unit. At the start of the expiration phase (downward travel of bellows) there shall be a definite blast of air at the atmosphere port of the quick discharge valve, accompanied by an initial acceleration of the bellows at the start of expiration. If acceptable, a mechanical stop shall be inserted between the bellows and the cycling unit lower plate, part 18, to prevent actuation of the lower bleeder valve, thereby causing the bellows to stop cycling. Soap solution shall be applied to the atmospheric port of the quick discharge valve. Any leakage at this port shall indicate a non-acceptable valve. If acceptable, soap solution shall be wiped off and mechanical stop removed from cycling unit.

4.4.5 Performance test, cycling unit and casualty circuit. The cycling unit complete with casualty circuit shall be subjected to the following tests:

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(a) Pressure regulator adjustment. - The contractor shall furnish a "T" adapter, to receive 1/8" nylon tubing at the through ports, and an accurate pressure gauge at the branch port, which shall be capable of easy installation and removal from the cycling unit between the pressure regulator secondary outlet and the input port of the three way valve. The cycling unit shall be operated from a power source (3.3.2.1.5.1) and the gauge observed during the downward travel of the bellows. The setting of the regulator shall be as specified (3.3.2.1.5.2) and during the downward travel of the bellows there shall be no fall in secondary pressure, which would indicate a leaky cycling unit component or faulty regulator.

(b) Bellows volume output. - The output of the bellows on inspiration shall be measured at the mask output port of the non-rebreathing valve by means of an anesthesia ventilation meter to determine compliance with the specification (3.3.2.1.5.5).

(c) Inspiration-expiration times. - The procedure for setting the inspiration-expiration times (3.3.2.1.5.6) shall be duplicated with respect to load on the casualty circuit, ambient temperature, and timing method employed. The inspiration and expiration times shall be recorded to 1/100 of a second, and the average of five (5) readings used to determine compliance with the specification requirements.

(d) Pressure setting. - The procedure for setting the pressure (3.3.2.1.5.7) shall be duplicated with respect to load on the casualty circuit and ambient temperature. The reading of the cycling unit pressure gauge shall be recorded that is required to initiate pressure cycling of the unit, and this reading shall be used to determine compliance with the specification requirement.

(e) Obstruction frequency. - The cycling unit shall be operated with the mask outlet port of the non-rebreathing valve completely blocked. A rubber stopper may be employed. The cycling frequency and the maximum reading of the cycling unit pressure gauge shall be recorded and used to determine compliance with the specification requirement.

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(f) Non-rebreathing valve resonation (pressure cycling). - The cycling unit shall drive a load as specified for checking pressure setting, (d) above. An audible resonating sound shall occur in the non-rebreathing valve. The fall-off of the cycling unit pressure gauge shall be observed to determine compliance with requirement specified in 3.3.2.1.5.7.

(g) Non-rebreathing valve unimpeded exhalation (volume cycling). - The cycling unit shall drive a load as specified for test (f) above except that the capacitive load shall be a rigid container of approximately 5 gallon capacity such that volume cycling occurs at a maximum cycling unit pressure gauge reading of from 40 to 50 cm H_2O pressure. During such operation the cycling unit pressure gauge shall indicate a rapid, complete drop-off of pressure at the start of expiration and the non-rebreathing valve shall exhibit a very low level of audible noise due to valve resonance.

4.4.6 Compressor capacity. Each compressor shall be tested for adequate capacity at the compressor manufacturer's plant. The compressor output shall be connected to a rigid metal tank having a minimum capacity of 100 ml. by means of a maximum of 2 feet of 3/16 inch inside diameter tubing. The tank shall be equipped with an accurate pressure gauge and an orifice 0.060 inches long and 0.043 inches in diameter. The compressor shall be driven at a speed of 2100 R.P.M. The compressor output shall be of such capacity that when equilibrium conditions are established in the tank that the pressure gauge reading shall be 15 psig, minimum.

4.4.7 Air compressor performance. The air compressor of the resuscitator shall be used to drive a calibrated cycling unit operating into various loads as specified in 4.4.5. The compressor shall be operated on 60 cycle A.C. power using the A.C. cord, and on line voltages (set by means of a suitable variac) of 105, 115, 125 volts and performance of cycling unit recorded with respect to inspiration-expiration times, pressure setting and obstruction frequency. The compressor shall then be operated on 24 volts D.C. (obtained by the series connection of two 12 volt storage batteries that are in good charged condition). The compressor shall only function when the proper polarity is maintained. If compressor operates when connections to battery are reversed, a faulty blocking diode is present which should be replaced. The performance of the cycling unit, as specified above, shall be recorded for D.C. operation. The values obtained with the air compressor shall be compared with those previously obtained using source specified in 3.3.2.1.5.1 to determine compliance with requirements.

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4.4.8 Radio interferance. The resuscitator electrical system shall be tested for compliance with the radiated and conducted electromagnetic emmission limits in accordance with MIL-STD-461 for class IIB equipment.

4.4.9 Rain test. The resuscitator unit, complete with all accessories, packed in the carrying case in accordance with 3.3.8 shall be subjected to a precipitation (rain) condition in accordance with MIL-E-5272, except a simulated rainfall of 3 ± 1 inch per hour for a period of five minutes, with a negligible leakage of water into the case. There shall be no evidence of damage to any component and there shall be no change in operation of the unit as result of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One resuscitator, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each resuscitator, along with component parts, shall be positioned and secured within their respective compartments in the carrying case. Components shall be so positioned as to prevent shifting of or damage to the contents. The unit shall be further packaged in a carton of appropriate size constructed in accordance with PPP-B-636, type CF, class weather-resistant. Closure and waterproofing shall be in accordance with the appendix of PPP-B-636.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type. Closure and strapping shall be in accordance with the appendix of the box specification.

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5.2.2 Level B. Unless otherwise specified, no overpacking will be required for level B protection.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The resuscitator is a portable unit suitable for the treatment of chemical warfare casualties, as well as for general medical resuscitation. It is suitable for field use in either a contaminated or uncontaminated atmosphere. It is a completely automatic unit which has been designed to give safe, efficient ventilation to a wide range of respiratory casualties without any machine adjustments required by the operator during the entire procedure. The resuscitator can be used by a ventilator of the controller type for use by an anesthesiologist. The resuscitator should not be used on children below 5 years of age.

6.2 Ordering data. Procuring documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Government furnished material (see 3.4).

* 6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog: FSN 6515-926-9157, Resuscitator, Field, 110 Volt, 60 cycle, AC, or 24 Volt, DC. It does not include all types, sizes or grades as indicated by the title nor all types which are commercially available.

6.4 The reference to vendors specified in the Drawing and Bill of Material, is furnished as a possible source and is intended as a guide.

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6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.6 Supersession data. This specification includes the requirements of Purchase Description No. 1 dated 8 July 1968.

Custodians:
Army - MD
Navy - MS
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Preparing activity:
Defense Supply Agency - DM

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5. PROBLEM AREAS

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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